

Problem 5

Moshood A. Garuba

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Problem 5.1 to 5.4 – Polynomial Regression Transformation

Given model:

$$y_i = b_0 + b_1 x^* + b_{11} (x^*)^2 \quad \text{where } x^* = x - \bar{x}$$

Expanding:

$$\begin{aligned} y_i &= b_0 + b_1(x - \bar{x}) + b_{11}(x - \bar{x})^2 \\ &= b_0 + b_1(x - \bar{x}) + b_{11}(x^2 - 2x\bar{x} + \bar{x}^2) \\ &= b_0 - b_1\bar{x} + b_{11}\bar{x}^2 + b_1x - 2b_{11}x\bar{x} + b_{11}x^2 \\ &= (b_0 - b_1\bar{x} + b_{11}\bar{x}^2) + (b_1 - 2b_{11}\bar{x})x + b_{11}x^2 \end{aligned}$$

Rewritten model:

$$y_i = b'_0 + b'_1 x + b'_{11} x^2$$

Answers:

$$5.1) \ b'_0 = b_0 - b_1\bar{x} + b_{11}\bar{x}^2$$

$$5.2) \ b'_1 = b_1 - 2b_{11}\bar{x}$$

$$5.3) \ b'_{11} = b_{11}$$

$$5.4) \ \text{Transformed Model: } y_i = (b_0 - b_1\bar{x} + b_{11}\bar{x}^2) + (b_1 - 2b_{11}\bar{x})x + b_{11}x^2$$

Bonus Question

For the GLM and more precisely the logistic regression model, write down an example for the link function and report the LOGIT model.

Consider a binary outcome variable $Y \in \{0, 1\}$, and let x be a single predictor variable. We define the probability of success (i.e., $Y = 1$) as:

$$\pi(x) = \mathbb{P}(Y = 1 \mid x)$$

1. Link Function (Logit Form)

In logistic regression, the logit link function connects the linear predictor to the probability $\pi(x)$:

$$\log \left(\frac{\pi(x)}{1 - \pi(x)} \right) = \beta_0 + \beta_1 x$$

2. Logistic Regression Model

Solving the equation for $\pi(x)$, we obtain the **logistic function**:

$$\pi(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$

3. Summary

The logistic regression model with one predictor in both forms:

$$\textbf{Logit form:} \quad \log \left(\frac{\pi(x)}{1 - \pi(x)} \right) = \beta_0 + \beta_1 x$$

$$\textbf{Exponential form:} \quad \pi(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$